# Gaussian Processes Changepoint Detection in Finance

## Final Presentation

T. Trivedi[1] Y. Kuang[2] C. Chen[3]

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2. Experiments and Results

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# What is changepoint detection used for?

Momentum family of investment strategies

- $r^{MOM}_{t,t+\delta} = sign(r^{INST}_{t-252\delta,t}) * r^{INST}_{t,t+\delta}$
- Used by all sorts of traders
  - used by high frequency traders on a microsecond basis
  - by intraday traders on a minutely basis
  - by hedge funds on a daily-weekly basis
  - by investment managers on a monthly basis

**Cumulative return of momentum strategies on currency universes**

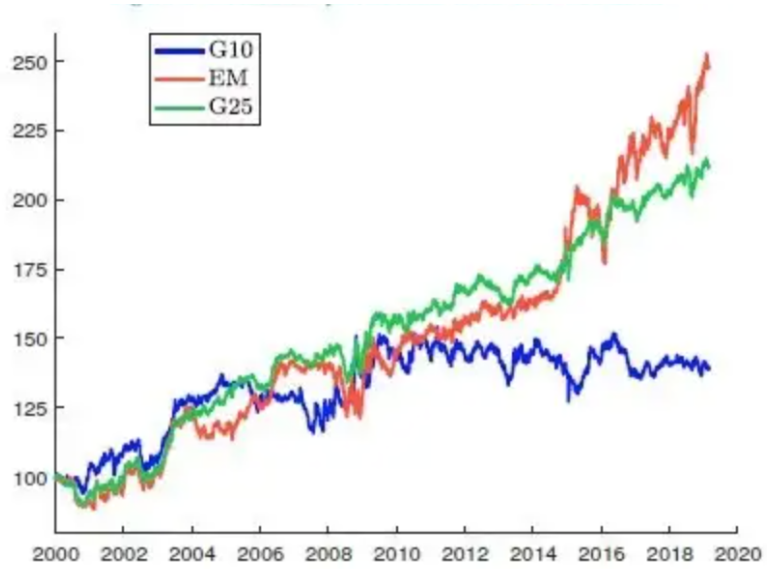


Figure: Baku et al, 2019

# Changpoint Detection (CPD)

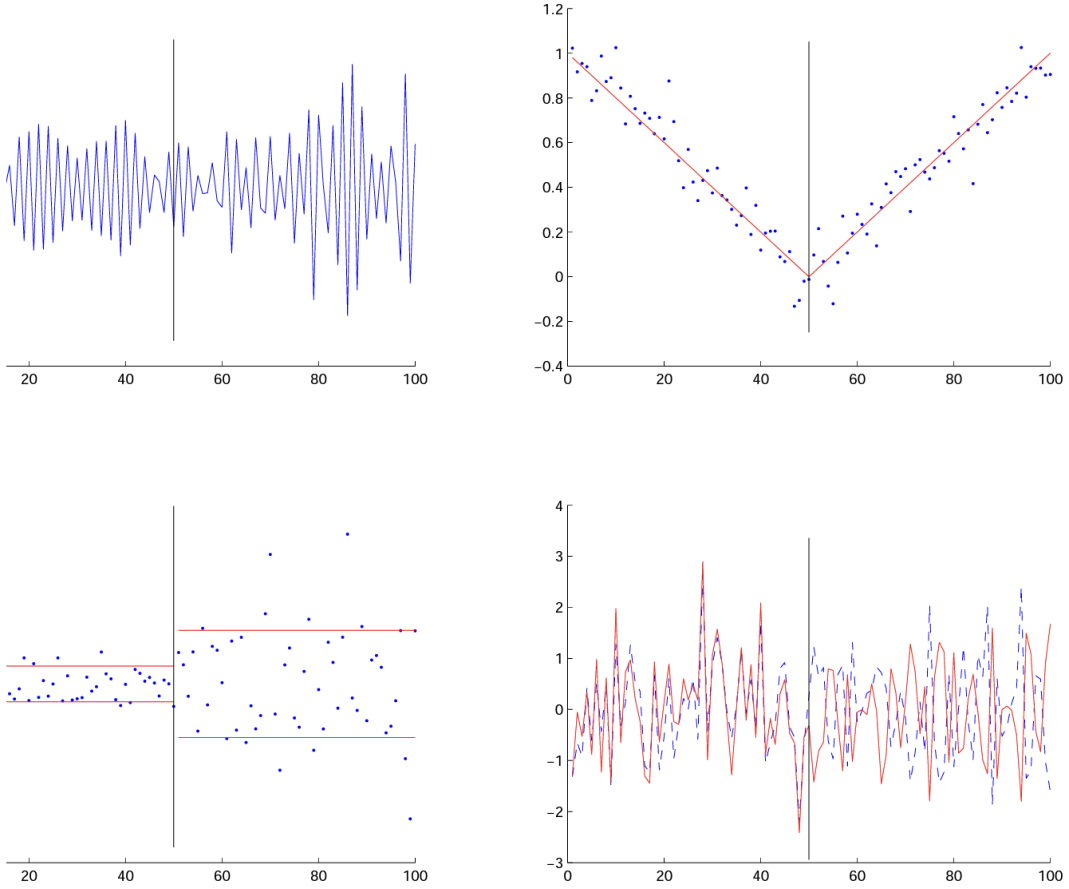


Figure: Various types of changepoints[1]

[1] Xuan and Murphy, "Modeling changing dependency structure in multivariate time series".

# Gaussian Processes for CPD

## Definition

**Changepoint (CP)** Consider a time sequence $\{x_i\}_1^n$ and changepoints $\{c_j\}_1^m$. Then, $\{x_i\}_{c_i}^{c_{i+1}}$ and $\{x_i\}_{c_{i+1}}^{c_{i+2}}$ are sampled from different distribution.

[2] Wood, Roberts, and Zohren, "Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection".

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Naive Kernel:

$$k_{\xi_M}(x, x') = k(x, x') \tag{1}$$

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$$k_{\xi_C}(x, x') = k_{\xi_1}(x, x')\sigma_C(x, x') + k_{\xi_2}(x, x')\bar{\sigma_C}(x, x') \tag{2}$$

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The hyperparameters $\xi_C = \{\xi_1, \xi_2, c, s, \sigma_n\}$ are optimized by the GPFlow

[2] Wood, Roberts, and Zohren, "Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection".

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# Dataset

The dataset includes the price of 9 currencies. The currencies are:

| Currency | Acronym |
|---|---|
| Australian Dollar | AUD |
| Canadian Dollar | CAD |
| Swiss Franc | CHF |
| Euro | EUR |
| Great Britain Pound | GBP |
| Japanese Yen | JPY |
| New Zealand Dollar | NZD |
| Swedish Krona | SEK |
| United States Dollar | USD |

# Kernel selection

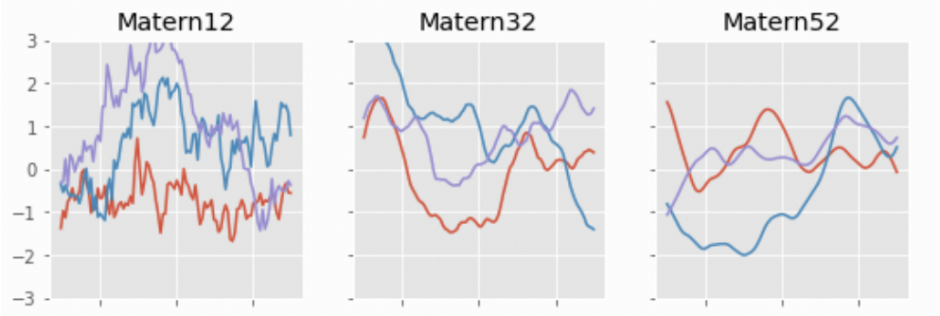



Figure: Matern Kernel Family from GPflow

Kernel Function for Gaussian Process
Motivation for Matern kernels?

- Related to Ornstein-Uhlenbeck processes, well studied Brownian motion model for financial time series
- Matern 1/2 Kernel
- Matern 3/2 Kernel
- Matern 5/2 Kernel
- Matern 1/2 + Matern 3/2
- Matern 1/2 + Matern 5/2
- Spectral Mixture Kernel[3]

[3] Wilson and Adams, "Gaussian process kernels for pattern discovery and extrapolation".

# Prior Training on Multi-currencies + Finetune

Since currencies are interdependent on each other, finding the changepoints on all currencies as a vector can utilize the structural dependency.

Then, we finetune the parameter for each currency from this pretrained hyperparameters.

We call this method "Prior Training on Multi-currencies + Finetune"

# Experiment Setup

CPD based trading strategy:

1. Start with a position $P = sgn(M)$ where M is the average return of the currency over a 90d window horizon and an empty list $P_{list} = [], R_{curr} = []$
2. Loop through the dates, recording new changepoints generated by the GP model at each step
   1. At each changepoint, flip the sign of $P$
      - Accept changepoint if score is above some threshold $\nu$ (0.98 in the results shown) where $\nu_t^{(i)} = 1 - \frac{1}{1+e^{-\left(\text{nlmn}_{\xi_C} - \text{nlmn}_{\xi_M}\right)}}$ and two negative marginal log likelihoods are of the naive and changepoint kernel
   2. Append $P$ to $P_{list}$; append the revenue to $R_{curr}$
3. $R = P_{list} \cdot R_{curr}$ where $R$ represent the revenue operated by the strategy solely based on the detected changepoint information.

Evaluation:

- Information ratio $= \frac{\mu_r}{\sigma_r}$ where $\mu_r$ is the annualized mean return of the strategy and $\sigma_r$ is the annualized standard deviation of return
  - Signal to noise ratio

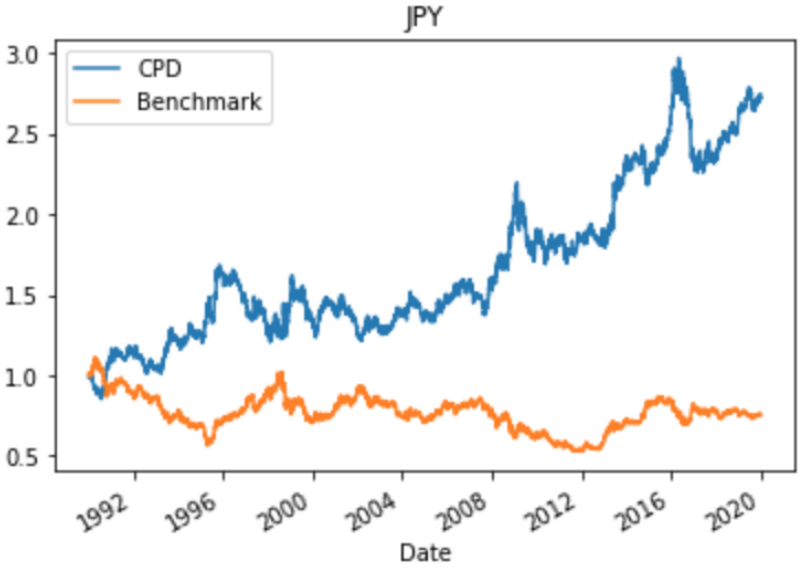
# Results- tables

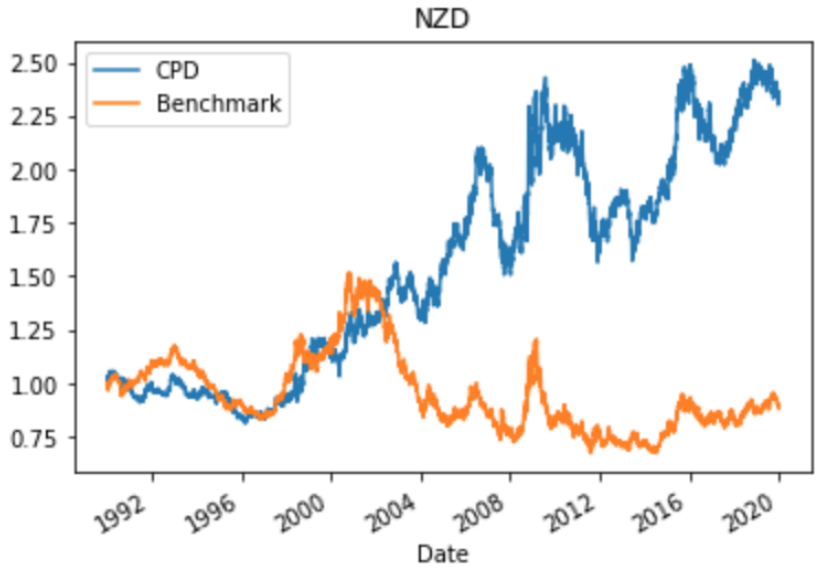
| Family of Kernel | Kernel Settings | Currencies Outperforming | Max Kernel IR | Max Baseline IR | Number of CPDs |
|---|---|---|---|---|---|
| Basic Kernel | 1/2 | 4 | 0.34 | 0.17 | 381 |
| | 3/2 | 3 | 0.26 | 0.17 | 402 |
| | 5/2 | 4 | 0.26 | 0.17 | 453 |
| Composition of Kernels | 1/2 + 3/2 | 2 | 0.32 | 0.17 | 420 |
| | 1/2+5/2 | 7 | 0.36 | 0.17 | 427 |
| Spectral Mixture | 5 Kernels | 4 | 0.17 | 0.17 | 2232 |
| | 10 Kernels | 0 | 0 | 0.17 | 2231 |
| Prior Training on Currencies | 1/2 | 2 | 0.33 | 0.17 | 412 |
| | 3/2 | 4 | 0.22 | 0.17 | 421 |

*Table 1.* Results of G9 Currency Experiment

Maximums and outperformance taken over lookbacks and currencies.

# Results- plots



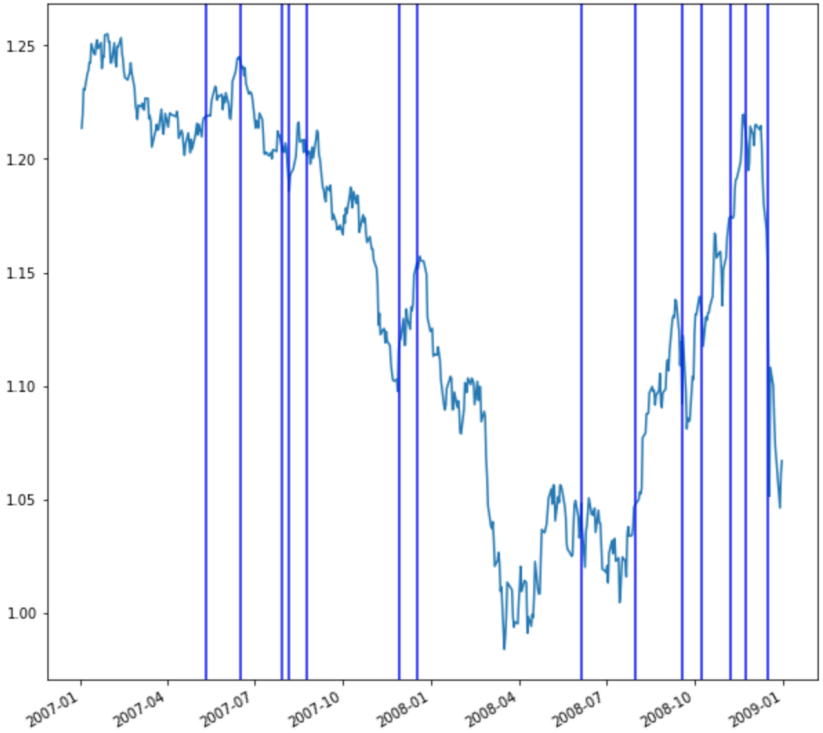




Figure: Change points on Swiss Franc Through GFC

# Future Work

- Test our approach on currency data from 2020-2022 under the COVID Market Crash
- Run performance as a confusion matrix of position vs avg return (performance might be biased by a few good trades as opposed to consistent wins)
- Compare the learned Spectral Mixture kernel to the Matern kernel

# Reference


- Wilson, Andrew and Ryan Adams. "Gaussian process kernels for pattern discovery and extrapolation". In: *International conference on machine learning*. PMLR. 2013, pp. 1067–1075.
- Wood, Kieran, Stephen Roberts, and Stefan Zohren. "Slow Momentum with Fast Reversion: A Trading Strategy Using Deep Learning and Changepoint Detection". In: *The Journal of Financial Data Science* 4.1 (2022), pp. 111–129.
- Xuan, Xiang and Kevin Murphy. "Modeling changing dependency structure in multivariate time series". In: *Proceedings of the 24th international conference on Machine learning*. 2007, pp. 1055–1062.